Big Data Computing

Written test 13/07/2022
(SOLUTIONS of PART2)

Exercise 1 Let P be a set of points and let $S = \{c_1, c_2, \ldots, c_k\} \subset P$ be the *optimal solution* to *k-center with z outliers* (the problem tackled by Homeworks 2-3). Call Z_S the set of outliers, that is the set of z points of P farthest from S and, for $1 \le i \le k$, define C_i as the set of points of $P - Z_S$ closest to c_i (ties broken arbitrarily). Thus, the optimal value of the objective function is:

$$\Phi^{\text{opt}} = \max_{x \in P - Z_S} d(x, S) = \max_{1 \le i \le k} \max_{x \in C_i} d(x, c_i).$$

Given an arbitrary set X of $k + z + 1$ points of P:

1. Show that there must exist a C_i that contains at least 2 non-outlier points of X (i.e., at least 2 points of $X \cap (P - Z_S)$).
2. Let $d_{min}(X)$ be the minimum distance between any 2 points of X. Using (a), show that $d_{min}(X) \le 2\Phi^{\text{opt}}$.
3. Based on the claim of (b), derive a bound on the initial guess r_{min} used by the sequential algorithm for k-center with z outliers implemented in Homework 2.

Solution.

1. X must contain at least $|X| - z = k + 1$ points of $P - Z_S$. Since the C_i's partition $P - Z_S$ into k subsets, there must exist a C_i containing at least 2 points of $X \cap (P - Z_S)$.
2. Let p, q be two points of $X \cap (P - Z_S)$ belonging to the same subset C_i, which must exist by virtue of (a). Clearly,

   $$d_{min}(X) \le d(p, q) \le d(p, c_i) + d(c_i, q) \le 2\Phi^{\text{opt}},$$

   where inequality $d(p, q) \le d(p, c_i) + d(c_i, q)$ derives from the triangle inequality.
3. The sequential algorithm for k-center with z outliers implemented in Homework 2, sets the initial guess r_{min} as $d_{min}(X)/2$, where X are the first $k + z + 1$ points of the input dataset P. Since the previous point holds for an arbitrary subset $X \subseteq P$ of $k + z + 1$ points, we have that $r_{\text{min}} = d_{min}(X)/2 \le \Phi^{\text{opt}}$

□

Exercise 2 Let T be a dataset of transactions over a set of items $I = I_1 \cup I_2$, where I_1 and I_2 are disjoint. An itemset X is said to be *fair* if it has even length and its items are half from I_1 and half from I_2. For a given support threshold minsup, let F_{2k} be the set of *frequent fair itemsets* of length $2k$. Two itemsets $X, Y \in F_{2k}$ are *compatible* if they have in common exactly $k - 1$ items of I_1 and $k - 1$ items of I_2. For $k \ge 2$, define

$$C_{2k+2} = \{X \cup Y \ : \ X, Y \in F_{2k}, \text{ such that } X \text{ and } Y \text{ are compatible}\}$$

Show that $F_{2k+2} \subseteq C_{2k+2}$.

Solution. Let $Z = Z[1]Z[2]\cdots Z[2k+2]$ be an arbitrary itemset of F_{2k+2} and, w.l.o.g., assume that $Z[i] \in I_1$, for $1 \leq i \leq k+1$, and $Z[i] \in I_2$, for $k+2 \leq i \leq 2k+2$. Define the following two itemsets

$$\begin{aligned} X &= Z - \{Z[k+1], Z[2k+2]\} \\ Y &= Z - \{Z[k], Z[2k+1]\} \end{aligned}$$

Note that $Z = X \cup Y$ and $|X| = |Y| = 2k$. By definition, both X and Y are subsets of Z and contain k items of I_1 and k items of I_2 (i.e., they are fair), hence $X, Y \in F_{2k}$. Moreover, X and Y are compatible since they share exactly $k-1$ items of I_1 $(Z[1]Z[2]\cdots Z[k-1])$ and $k-1$ items of I_2 $(Z[k+2]Z[k+3]\cdots Z[2k])$. Therefore, $Z = X \cup Y$ must be included in C_{2k+2}. $\square$